

Coding & Solution

Date	05 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/EjjipurapuKavya/AI-ML-and-GEN-AI-Track-Project-Template-Kavya
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-learn

Key Features Implemented	• Crop Recommendation using Machine Learning • Soil Nutrient Analysis (N, P, K) • Weather-based Prediction (Temperature, Humidity, pH, Rainfall) • Flask Web Application • Responsive User Interface
Pending / Incomplete Features	• Fertilizer Recommendation Module (Future Enhancement) • Crop Yield Prediction Module (Future Enhancement) • Weather API Integration
Setup / Run Instructions	1. Install Python and required libraries (Flask, scikit-learn, pandas, numpy, joblib). 2. Download the project from GitHub. 3. Open the project folder in Command Prompt or VS Code. 4. Run python app.py. 5. Open http://127.0.0.1:5000 in a web browser. 6. Enter soil and weather parameters to receive crop recommendations.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	yes
2	Meaningful variable and function names are used	yes
3	Code includes comments / documentation where necessary	yes
4	Error handling is implemented for critical operations	yes
5	The application runs without critical errors	yes
6	Code is committed to a version control repository	yes

Additional Notes / Comments:

- ☐ The project uses a Machine Learning model trained on the **Crop Recommendation Dataset** from Kaggle.
- ☐ The solution is developed using **Python, Flask, HTML, CSS, and Scikit-learn**.

- The application provides fast and accurate crop recommendations based on soil nutrients and environmental conditions.
- The architecture is modular, making it easy to maintain and extend with additional agricultural services in the future.